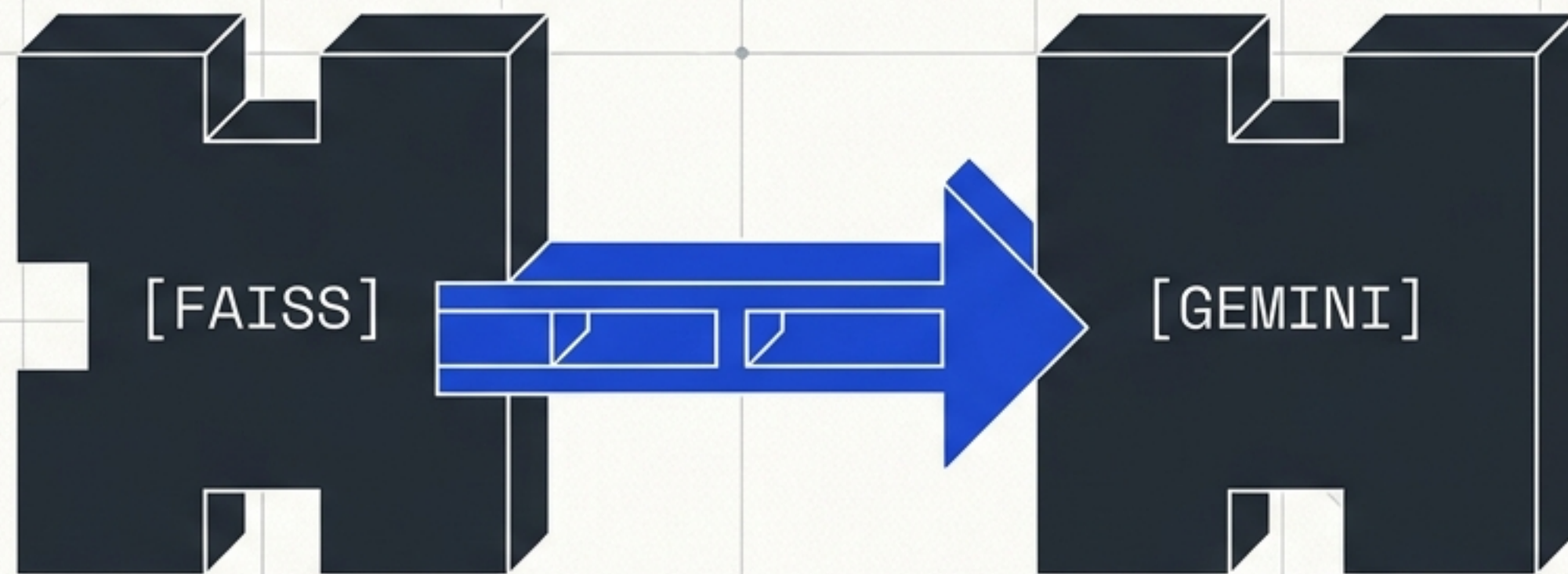
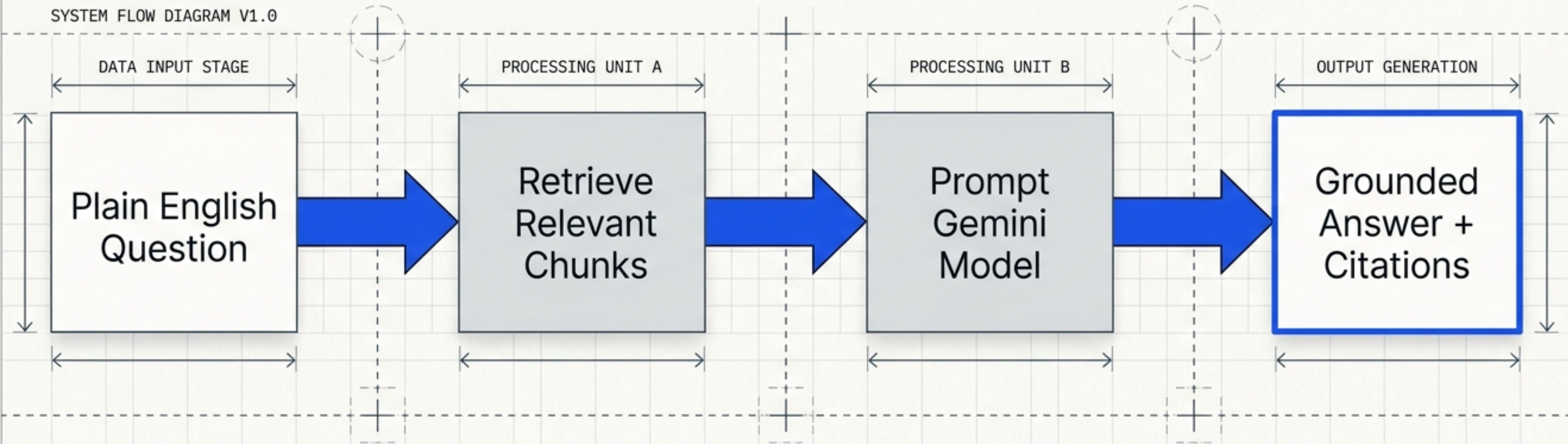


Building the RAG Core: Retrieval + Generation

HOLBERTON PHASE 2 //
ARCHITECTURE &
IMPLEMENTATION

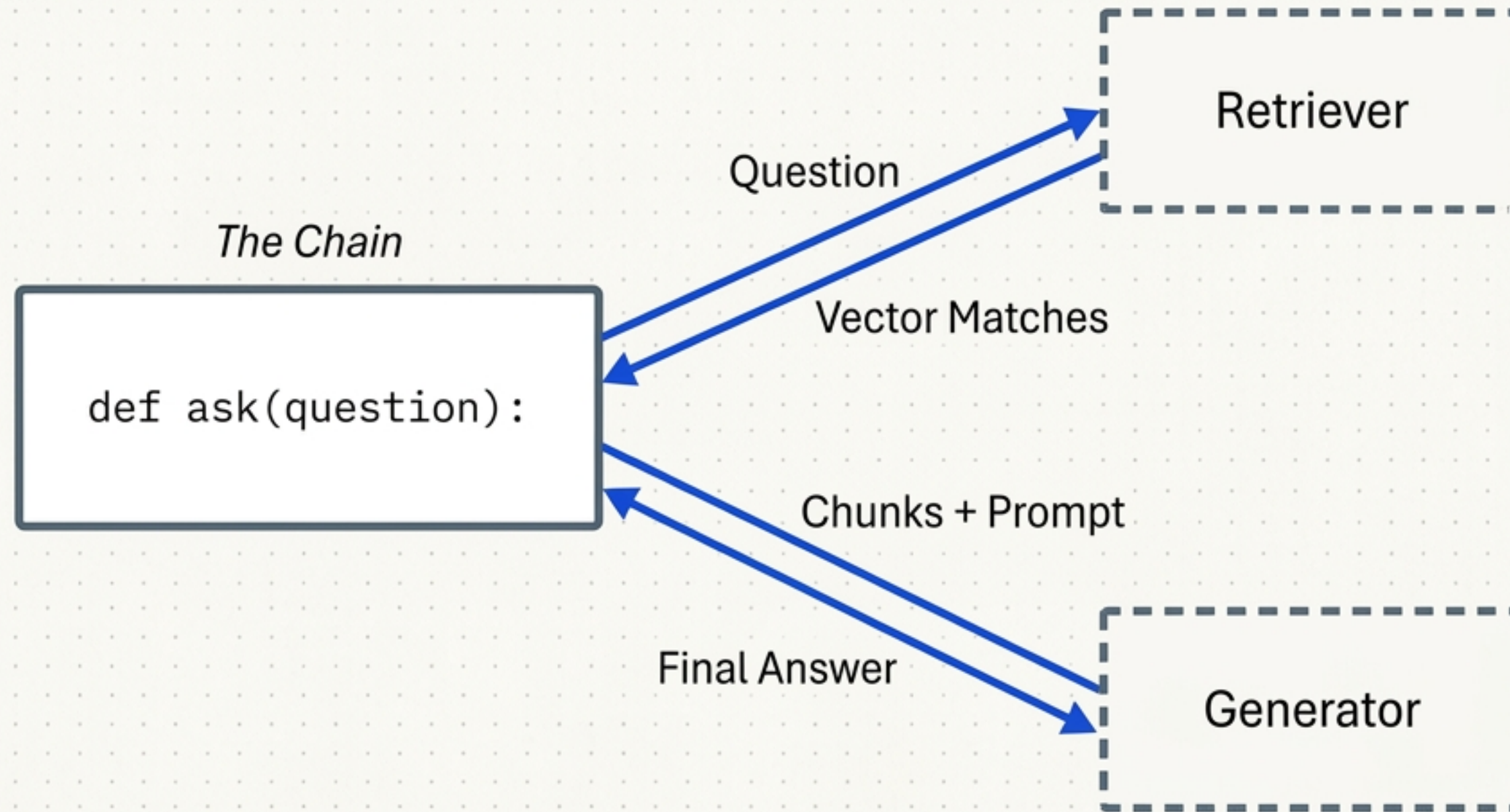


A Question Enters, A Grounded Answer Exits



This is the first time the system behaves like an AI assistant.
Everything runs **purely** in Python—no web interface yet.

The Component Ecosystem



The RAG chain is the single entry point. It wires the components together and is what Flask will call in Phase 3.

Strict Separation of Concerns

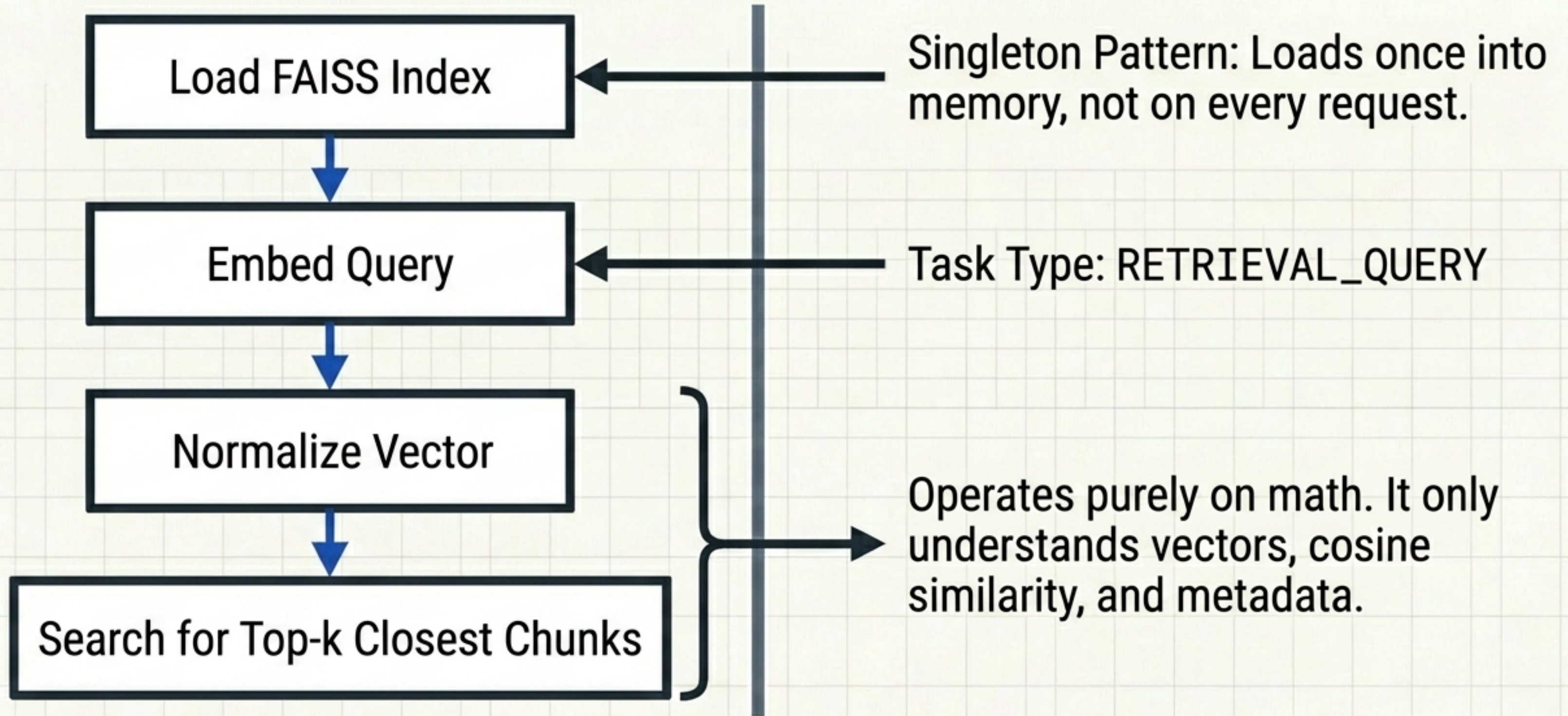
THE RETRIEVER

Core Job	Find semantic relevance
Input	Plain text question
Output	Top-k text chunks + metadata
Strict Rule	NEVER calls the Gemini LLM

THE GENERATOR

Core Job	Answer from context
Input	Retrieved chunks + user question
Output	Natural language + source list
Strict Rule	NEVER searches the FAISS index

Retriever Mechanics: **Search & Score**



The Anatomy of a Retrieved Source

Data Structure (Dict)

filename

title

content

score (0 to 1)

Query: How can I transfer?

CampusTransfers.md score: 0.7363

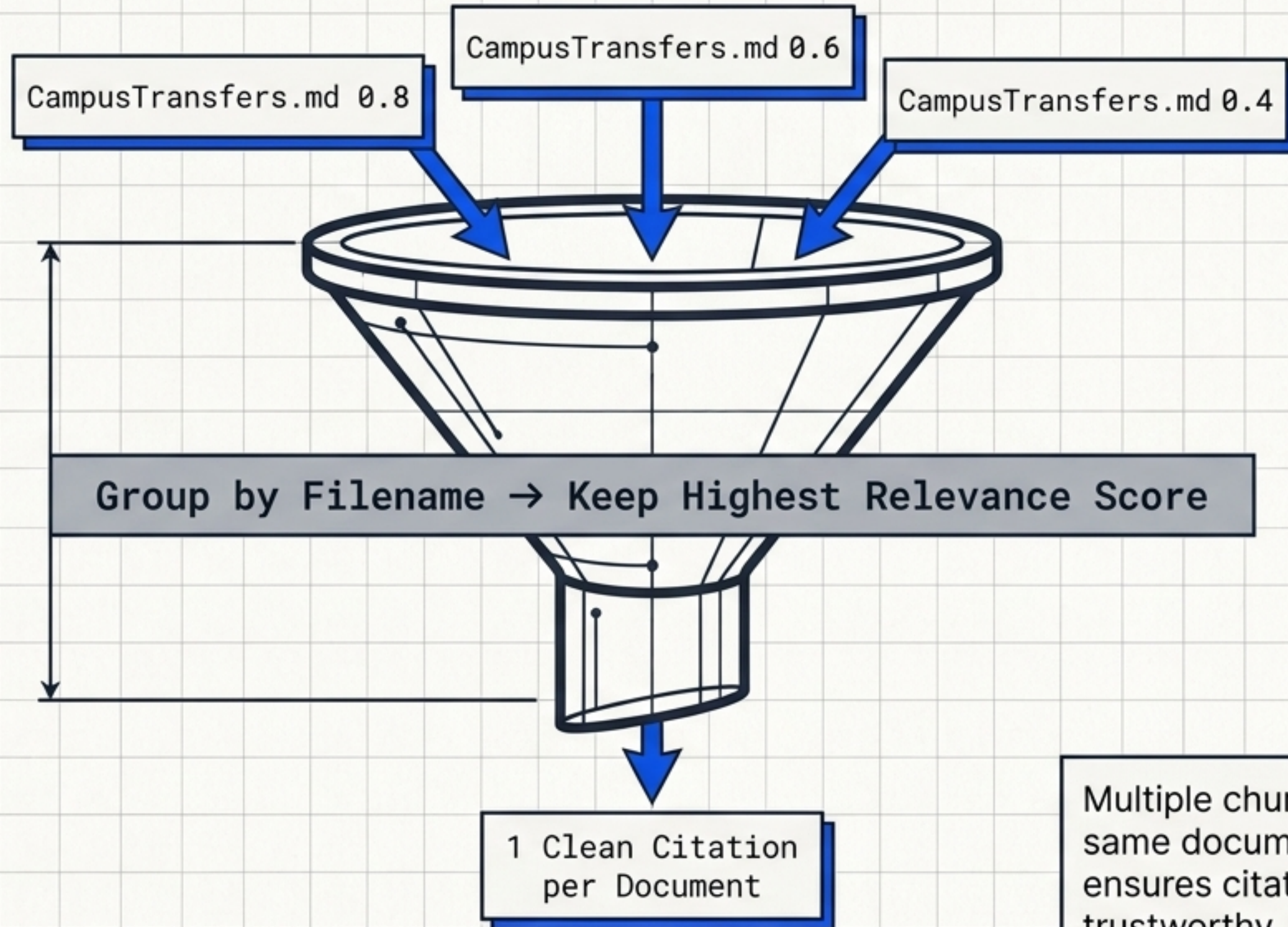
StudentTransfers.md score: 0.6697

StudentTransfers-FAQ.md score: 0.6569

Specializations-FAQ.md score: 0.6232

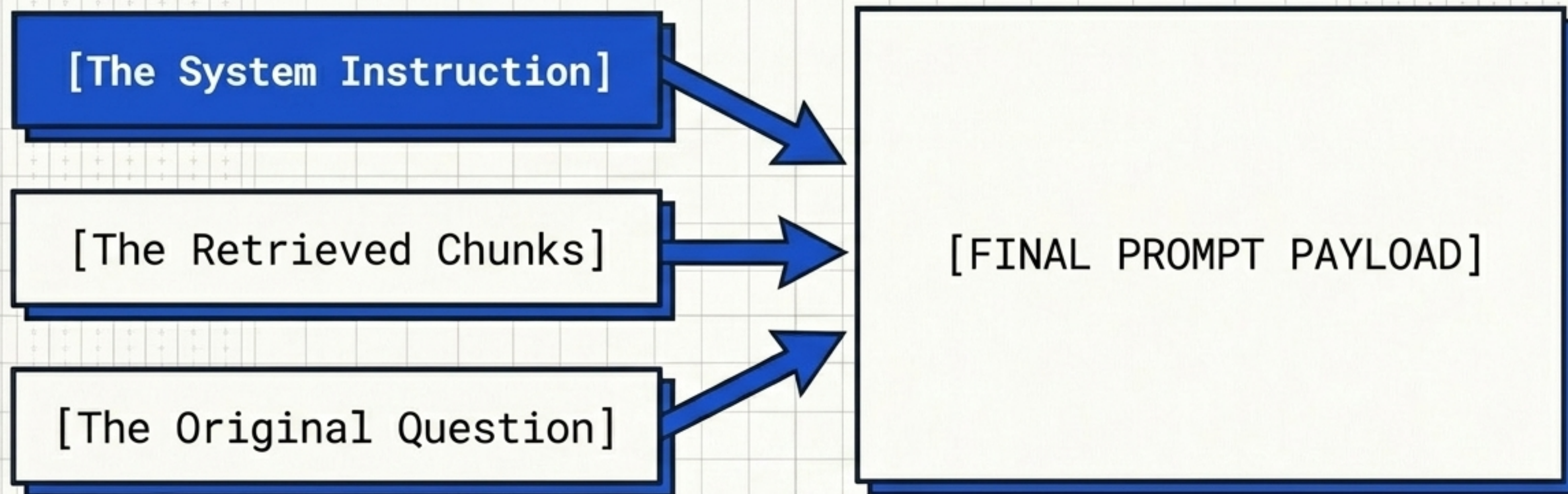
By default, the system returns the top 5 closest meanings (configurable via .env).

Source Deduplication



Multiple chunks often come from the same document. Deduplication ensures citations are clean, trustworthy, and readable.

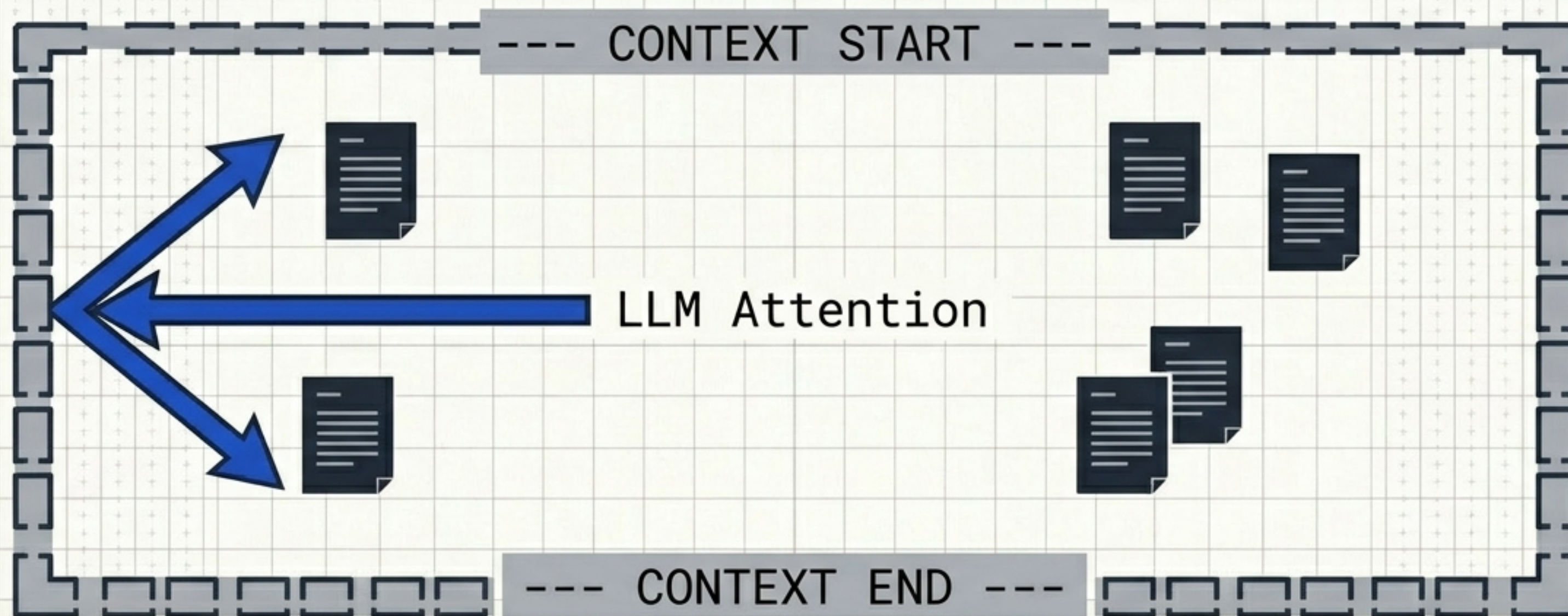
The Generator: Assembly & Constraint



Answer only from the context provided. If the context does not contain enough information, say clearly: I could not find a clear answer.

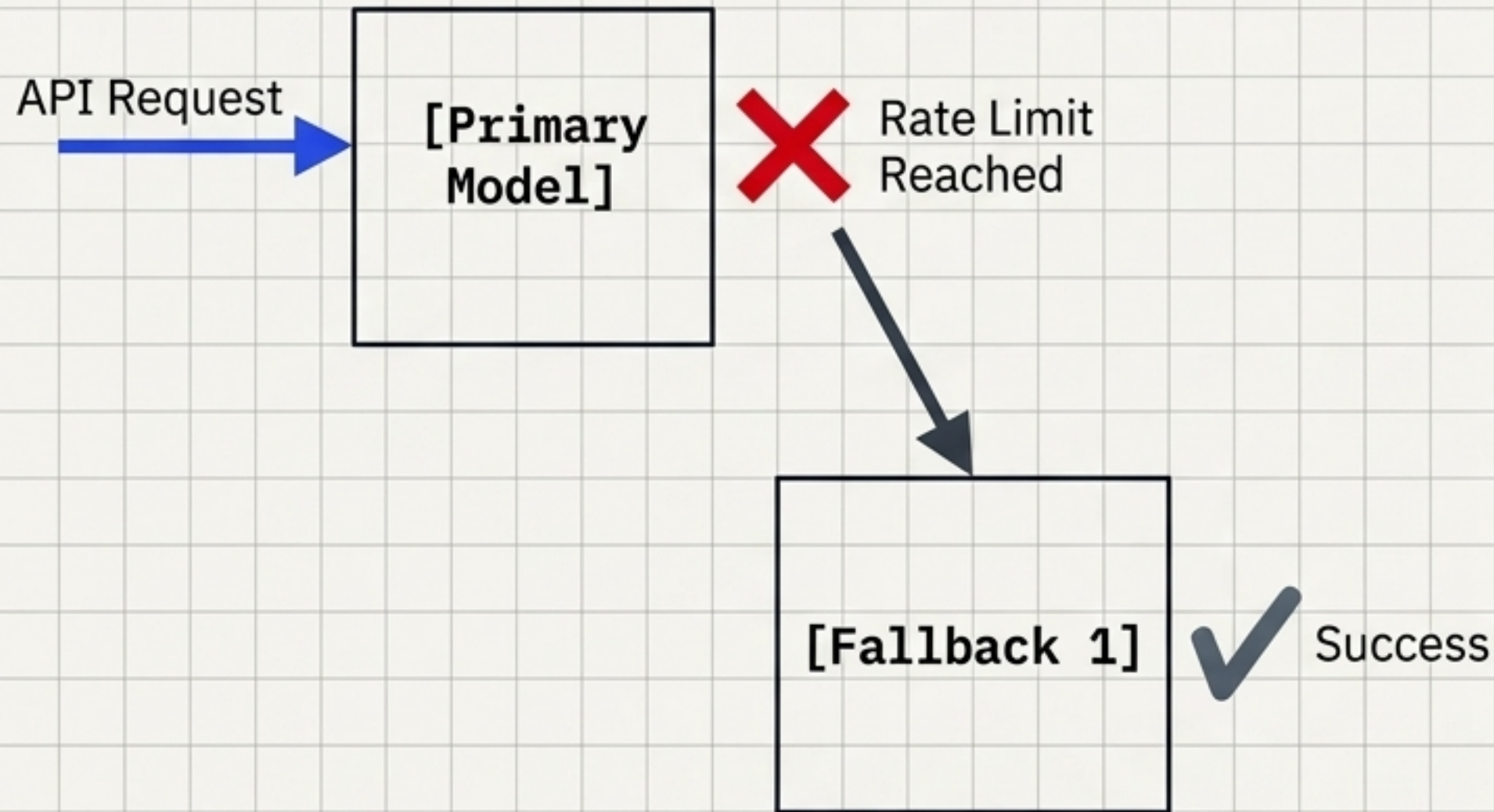
This single instruction is the primary defense against LLM hallucination.

Grounding Forces Conceptual Boundaries



Hallucination is plausible but incorrect information. Grounding locks the model inside specific boundaries it cannot cross.

The Waterfall Fallback Chain



TIER	DAILY LIMIT	RPM LIMIT
Primary: gemini-2.5- flash-lite	1000/day	15 RPM
Fallback 1: gemini-2.5- flash	250/day	10 RPM
Fallback 2: gemini-2.0- flash	100/day	5 RPM

Graceful degradation ensures students always get an answer, mirroring how production systems handle API limits.

Live Validation: The Loop is Closed

> Question 5: How are projects evaluated at Holberton?

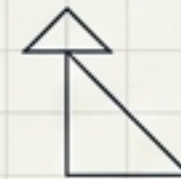
Retrieving sources...

ProjectScoring-Checker.md relevance: 0.7877

ProjectTypes.md relevance: 0.7854

ProjectScoring.md relevance: 0.7711

Generating grounded response...



Highly relevant sources (>0.74 score). The retrieval finds exactly the right documents; the generator synthesizes them.

Intentional Constraints (What Phase 2 Omits)

No Web UI

Answers print to terminal only.

No Flask Layer

The chain is pure Python, no HTTP.

No Conversation Memory

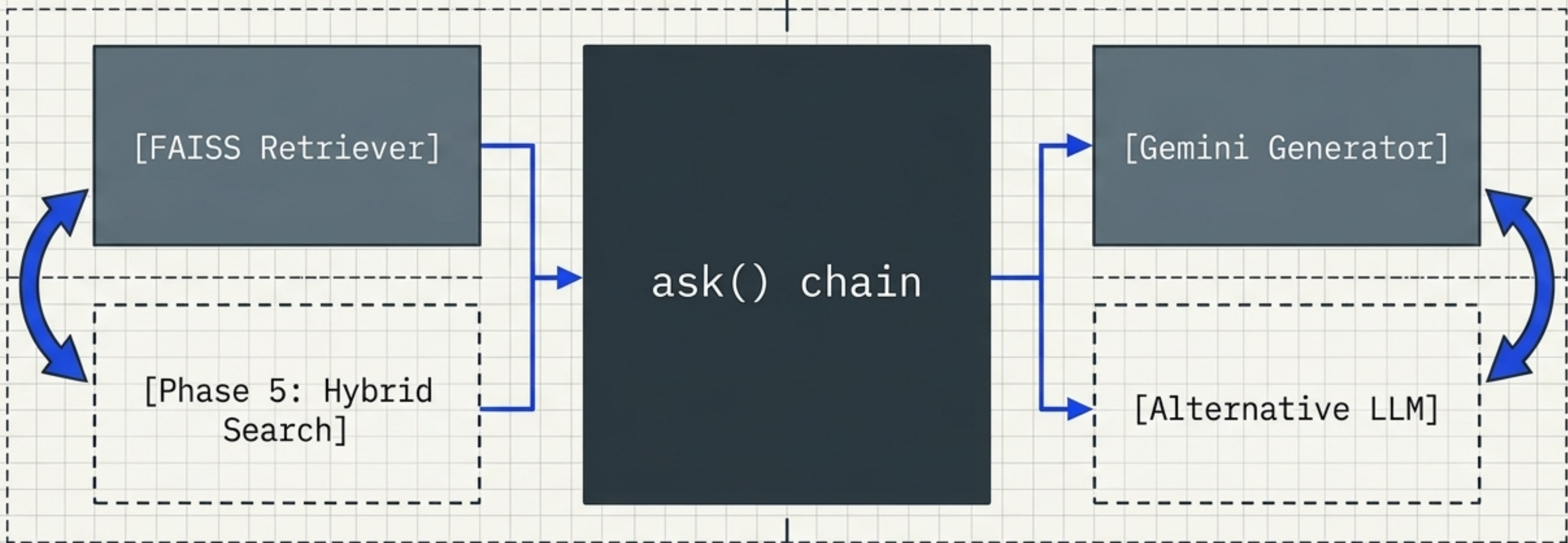
Each question is independent.

No Evaluation System

Quality measurement comes later.

Building layers separately allows each to be tested in isolation before adding the complexity of the next.

The Seam, Not The Logic



The chain does not know how to search or generate. It only knows how to connect. By keeping the Retriever and Generator ignorant of each other, the system becomes extensible and future-proof by design.

Architectural Truths of Phase 2

01

Grounding is Non-Negotiable

A system prompt constraining the model to retrieved context is the difference between a trustworthy tool and a liability.

02

The Chain is the Seam

Connect independent components without entangling them.
Extensibility is engineered, not accidental.

03

Retrieval Drives Quality

Answers are only as good as the chunks provided. In production, the hardest engineering effort goes into retrieval, not generation.